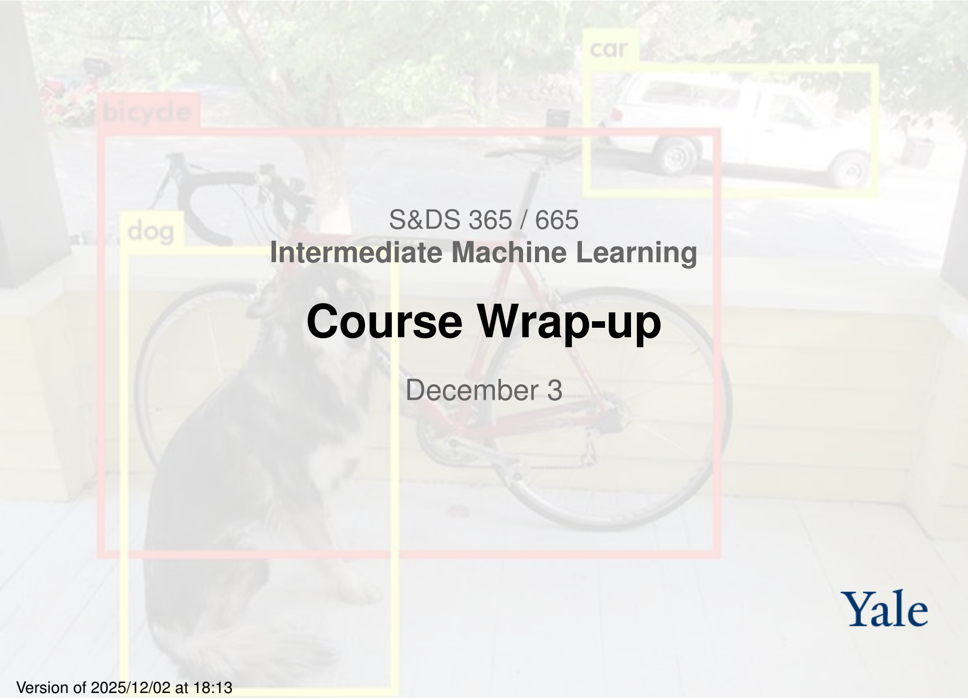


S&DS 365 / 665

Intermediate Machine Learning

Course Wrap-up

December 3

Yale

For today

- New directions in ML/AI
- Differing views of our AI future
- Final exam logistics
- IML Survivor

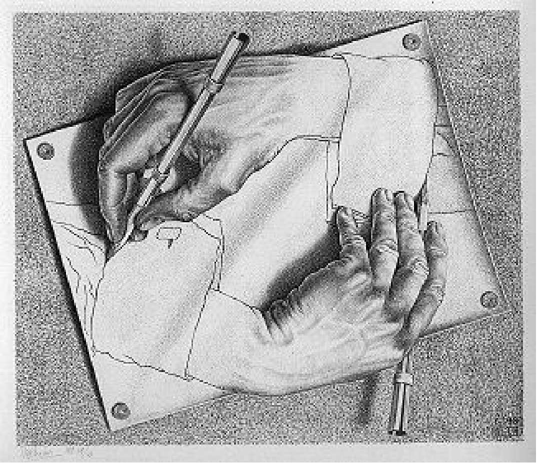
But first: The usual reminders

- Assn 5 due tomorrow
- Final exam: Friday Dec 12 at 9am
- Practice exams posted
- Review sessions:
  - Monday, December 8 at 5pm, KT 1101 (Zhehao)
  - Tuesday, December 9 at 5pm, Zoom (Andrea)
  - Wednesday, December 10 at 4pm, KT 1208 (John)

Frontiers of ML and AI

Some personal perspectives

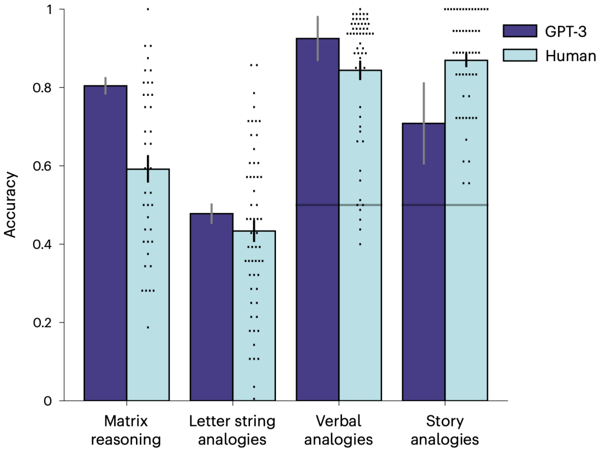
For today's discussion, recall the view that ML is the technological core that underpins AI systems. An analogy would be the distinction between nuclear physics and nuclear energy.



You scratch my back,
and I’ll scratch yours

$$[A, B] = AB - BA = 0$$

LLMs have emergent abstraction abilities



https://www.nature.com/articles/s41562-023-01659-w

But LLMs can fail catastrophically

"Two shapes are in a room. The shape next to the cube is blue. The shape next to the sphere is red. Which is red, the cube or the sphere?"

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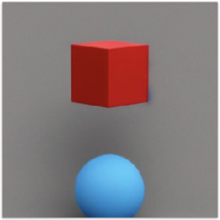

But LLMs can fail catastrophically

Based on the given information, it can be inferred that the sphere is red. The statement specifies that the shape next to the sphere is red, implying that the sphere itself is red. The color of the cube is not mentioned, so we cannot determine its color from the given information.

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Shortcomings are masked

- Innovations with Finetuning/RLHF hide these deficiencies
- System is trained to convince us
- Can lead to over-confidence and incorrect reasoning
- A central issue with “AI safety”

Lack of a world model

Two types of intelligence

1. "Sensory"— acquire semantic and procedural knowledge
2. "Relational"— rapid abstraction, association and generalization

Can both types be supported in a single architecture?
This is a frontier question for AI

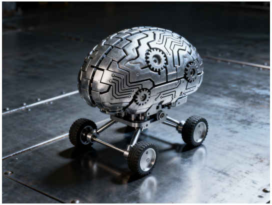
Supernatural intelligence?

What didn't evolution
discover wheels?

Supernatural intelligence?

What didn’t evolution discover wheels?

Will AI evolve “cognitive wheels”?



ARC-AGI Prize

arcprize.org; Chollet, “On the measure of intelligence” (2019)

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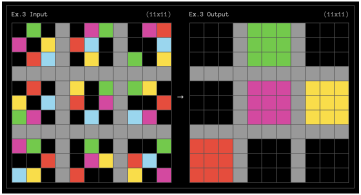
“Platonic” abstractions

- What is required to discover *useful* “cognitive wheels”?
- Platonic view of primitive abstractions (small natural numbers, spatial relations) — they are real and exist independent of human thought



- Algorithms should be encouraged (through inductive biases?) to employ these abstractions

Chollet, “On the measure of intelligence” (2019); Chollet, “OpenAI o3 Breakthrough High Score on ARC-AGI-Pub, 2024. `https://arcprize.org/blog/oai-o3-pub-breakthrough`, Beger et al. (2025) “Do AI models perform muman-like abstract reasoning across modalities?” arXiv:2510.02125

“Platonic” abstractions

“Using such abstractions is how we humans are able to understand our world, predict probable outcomes of our decisions, and to efficiently deal with novelty. These are all things we want AI systems to do in more trustworthy ways.”

Mitchell, “AI: A Guide for Thinking Humans,” October 2025; Beger et al. (2025) “Do AI models perform muman-like abstract reasoning across modalities?” arXiv:2510.02125

Creativity and rigorous reasoning

- We need new ML ideas to enable abstraction and analogy
- Humans often resist logical and evidence-based reasoning
- Can AI be based on more “rigorous” reasoning?

"We have Paleolithic emotions, medieval institutions and godlike technology. And it is terrifically dangerous, and it is now approaching a point of crisis overall."

Edward O. Wilson (2009)